



STATUS REPORT

Namibia — digital transformation and data governance status report

Edition of 2026-09-06 · compiled by Claude Opus from the documents in the Corpus repository

GOVERNANCE · FINANCE · ICT INFRASTRUCTURE · DPI · DIGITALISATION · TECHNOLOGY · CAPACITY · INCLUSION · DATA · GEOPOLITICS

Governance

Strategies, plans and policies

Namibia has no dedicated government entity responsible for data governance or data management, and no data governance strategy or policy (2025) — an absence underneath an unusually crowded shelf of published strategy.

The shelf is genuine. The Office of the Prime Minister's Digital Government Strategic Roadmap 2024-2026, "Digital First Services for All", carries the connectivity programme alongside the National Digital Strategy, and the sixth National Development Plan targets internet penetration of 90 percent by 2030. The Ministry of Information, Communication and Technology's Strategic Plan 2025-2030 sets the digital agenda, including a national data centre and measures to close the urban-rural divide. A National Digital Health Policy 2026-2036 was launched in February 2026, planning electronic medical records and broadband for every health facility, rural areas first. The headline commitments are to raise access to digital technology to 70 percent and to digitise 100 percent of government services by 2029/30.

Set against that, the World Bank's 2025 assessment finds no GovTech or digital transformation strategy, a whole-of-government approach only planned or in draft, and nothing published on progress or spending, no open source policy for the public sector and no national strategy on disruptive or innovative technologies, and, although a GovTech entity and a coordinating body do exist, nobody monitoring or reporting digital spending across government and no annual progress report from the GovTech institution.

The wider governance trend runs the same way: of the 96 indicators in the 2024 Ibrahim Index, Namibia improved on 35 between 2014 and 2023 and declined on 54 (2023), with public administration one of the largest contributors to the decline.

Delivery ownership is at least moving in payments, where licensing and supervision of payment service providers passes to the Bank of Namibia under a new Payments Association of Namibia framework, separating the industry association from the supervisory function it had sat close to (July 2026). On cyber, the national cybersecurity strategy has been opened to review and public submission, and Cabinet approved proposals in November 2025 for a government-sector incident response team and security operations centres, with the Office of the Prime Minister coordinating.

Legislation and regulation

Namibia's most consequential digital statute does not yet operate: the [Civil Registration and Identification Act 13 of 2024](#) was assented to on 13 December 2024 and gazetted on 30 December 2024 but had still not been brought into force as at 2026. It would supply the legal basis for the electronic identity card and for biometric capture and authentication, while the [Identification Act 21 of 1996](#) remains the governing law for the population register itself.

The framework law for electronic transactions is in a similar half-state. The [Electronic Transactions Act 4 of 2019](#) empowers the minister to issue guidelines on system interoperability but is only partly in force, Chapters 4 and 5 and section 20 having never been commenced, and no Namibian law requires one part of government to share data with another. A [Cybercrime Bill](#) was still in draft in January 2026 and still being finalised in July 2026, leaving enforcement to the [Electronic Transactions Act](#), the [Communications Act](#) and the [Penal Code](#); the critical information infrastructure designation it provides for does not exist in law, and the [Data Protection Bill](#) alongside it has been pending far longer.

What is in force is narrower. The [Access to Information Act 8 of 2022](#) gives Namibians a statutory right to request information held by public bodies, and the [Payment System Management Act 14 of 2023](#) gives the Bank of Namibia power to regulate and oversee the national payment system, approve its rules and issue directives.

Licensing is where statute has bitten hardest. CRAN refused Starlink a comprehensive service licence and the spectrum it sought, in a decision gazetted on 23 March 2026, finding the applicant met three of six statutory criteria; the failures were ownership, the company being nil percent Namibian-owned against a 51 percent requirement and having asked for no exemption, compliance history, having previously operated unlicensed, and national security and regulatory oversight. Section 31 of the [Communications Act](#) opened a 90-day reconsideration window; of 624 requests, 622 failed the legal threshold, so the refusal stood on 24 June 2026, with a fresh application already filed on 8 June.

Data protection

Namibia has no comprehensive data protection legislation and no data protection authority (2024), and the [Data Protection Bill](#) that would create both was still being finalised in July 2026, roughly twelve years after drafting began, having returned to parliament in 2025 unpassed. Nothing fills the gap: there is no entity monitoring compliance with data protection rules and nothing published on privacy complaints or their handling (2025), and no data localisation requirement and no regulation of cross-border personal data flows (2026).

What the drafts promise has no force. The Bill would establish a Data Protection Supervisory Authority, give individuals a right to know and access the personal data held about them, and restrict cross-border transfers to jurisdictions offering adequate protection (2022), and its 2020 text carried a full data-subject rights part, from access and erasure to a right not to be subject to automated decision-making. Separately, the uncommenced [Civil Registration and Identification Act 13 of 2024](#) would give every data subject access to their own record, require memoranda of agreement before civil-register data is shared onwards, require a court order for law-enforcement access, mandate an access register and create an Appeals Tribunal.

Collection has not waited. Biometric enrolment for the national identity system and SIM registration against a national ID or passport, mandatory since 2023 both run without a statute, and the Office of the Prime Minister names internal misuse of e-ID data as a larger risk than external attack. Losses are not hypothetical: [Telecom Namibia](#) lost 626GB of data to the [Hunters International](#) ransomware group in December 2024, exposing records belonging to eight ministries, five regional councils and ten municipalities, and the [Namibia Students Financial Assistance Fund](#) published the personal details of more than 7,000 students on its own website in October 2025, after which the head of the national incident response team called for urgent finalisation of the data protection law.

No general duty to notify follows, because only national payment system participants must report a successful cyberattack to the central bank within 24 hours. Data sovereignty is nonetheless available as an argument: CRAN rested part of its refusal of Starlink's licence on it, holding that 100 percent foreign ownership undermines jurisdiction and the enforceability of compliance obligations.

Regional collaboration

The furthest Namibia's digital estate reaches across a border is its identity card: the national ID has been accepted as a travel document for crossing into Botswana under a bilateral agreement in effect since 2023, so citizens move between the two countries without a passport.

Payments are the second channel. Namclear, the national clearing house, signed an agreement with PayInc in March 2026 to connect Namibia to TCIB, the SADC regional instant cross-border payment scheme, with low-value push payments to Common Monetary Area countries being introduced in phases.

Namibia's formal integration position is strong and improving: regional integration is one of its ten most improved governance measures, up 14.8 points over 2014-2023 to 64.7 out of 100 and 4th of 54 (2023), and it ranks 6th of 54 for overall governance on 63.9, well above the African average of 49.3, though its own score has fallen 3.3 points since 2014 while that average rose (2023).

Ambition is also being borrowed from the neighbourhood: the ICT minister said in June 2026 that Angola's sovereign data centre in Luanda had encouraged Namibia's own data-centre ambitions.

Standards

Namibia has no government interoperability framework, no data quality framework, no guidance for replacing legacy government information systems and no system for monitoring their uptime (2025), so public systems are specified one at a time rather than against a shared published baseline. The e-procurement portal is in use and publishes tender notices and contracts, but not to the Open Contracting Data Standard, and it exchanges no data with other government systems (2025).

Where published standards do exist, they are the payment sector's. The Namibia Interbank Settlement System has moved to the ISO 20022 messaging standard, a national QR payment standard, NamQR, has been settled and open banking specifications are complete (July 2026), though no go-live date, merchant count or participant list has been published for either NamQR or open banking.

The national trust anchor sits inside the communications regulator rather than a separate cyber body: CRAN states that it acts as Namibia's Root Certification Authority, anchoring the national digital trust framework and regulating digital-certificate and certification-service providers. That anchor acquired its first customer at the end of August 2026, when the regulator and a German identity vendor signed an implementation agreement with the home affairs ministry connecting the ministry to the national trust framework, the ministry becoming the first certification service provider through its issue of electronic identity documents. It has begun recruiting standards work outwards, signing a three-year memorandum of understanding with the Namibia Chamber of Commerce and Industry in Windhoek on 30 July 2026, overseen by a joint steering committee, covering public key infrastructure, cybersecurity, e-trade facilitation, regulatory standards and legislative reform.

Product standards are being written faster than public-sector ones. The national standards institution launched a strategy in October 2025 targeting 1,200 new standards in three years, against 235 published since its establishment in 2008, set beside 1,600 in Botswana, 6,500 in Zambia and more than 10,500 in South Africa, and a package covering food products and food safety management was adopted in June 2026. None of that programme reaches government information systems.

Public debate and participation in policymaking

Namibia has the freest media in Africa on the 2024 Ibrahim Index, 1st of 54 with a score of 83.1 out of 100, but that score has fallen 10.7 points since 2014 (2023) – the continental lead is being held on a declining measure. Digital freedom scores 79.6, 11th of 54, and has risen slightly over the decade (2023).

The channels the state itself provides are thin. There is no government platform for citizens to lodge complaints, suggestions or information requests about service delivery, no published service standards such as response times, and no evidence of services changing in response to feedback (2025), and no national online platform through which citizens or businesses can take part in policy decisions, submit petitions or give anonymous feedback (2025). The Access to Information Act 8 of 2022 supplies the right to ask, but nothing is published on how the law is being applied, including how many requests are received and granted (2025).

So the argument happens outside those channels, and in 2026 it has been about the electronic ID. Residents of the Omusati region protested in June 2026 on the claim that the card would microchip and track citizens, and the Minister of Home Affairs rebutted it in the National Assembly; social-media claims that the card exposes citizens' data followed, with the Ministry of Home Affairs rebutting them on the record and by name at Gobabis (August 2026); and undocumented Namibians petitioned in July 2026 for the rollout to be suspended until a years-long backlog of citizenship and identity documents is cleared, arguing that a digital layer on an incomplete foundational register would entrench exclusion from jobs, education and services.

Scrutiny of digital policy in parliament has come from the opposition benches: its asks over a Chinese-funded smart city are disclosure of the agreements, annexures, contractors, data architecture and lifetime operating costs, public hearings, independent security assessments, and a rights-respecting Data Protection Act enacted before the build proceeds (August 2026). The most recent draft of that Bill has not been published for the public to see.

Finance

MoUs and other agreements

CRAN and the Namibia Chamber of Commerce and Industry signed a three-year memorandum of understanding in Windhoek on 30 July 2026, covering public key infrastructure, cybersecurity, e-trade facilitation, regulatory standards and legislative reform, and put a joint steering committee over it. The committee is the part worth watching: a memorandum of this kind fixes what the parties will work on and commits them to work on it, not to spend or to deliver by a date, so a standing body is what turns the signature into work.

Namibia and China signed a Smart City arrangement at the end of June, with the pilot forming part of the Windhoek People-Centred Smart City Strategy (2026-2036), a decade-long programme to modernise the capital (2026). It is the one of these agreements with funding named alongside it.

Two more were signed on cybersecurity. Namibia and India agreed a cybersecurity memorandum at State House, between President Nandi-Ndaitwah and Prime Minister Modi, alongside cooperation on UPI-based payments and digital governance (July 2025), and the Namibia Institute of Public Administration and Management signed with the Global Cybersecurity Centre on 4 June 2026 to train public servants in cybersecurity.

New investments

The largest money going into Namibia's digital estate is a private operator's own: Paratus put NAD 600 million, about US\$33.7 million at 2025 rates, into building its new Namibian mobile network (2025), roughly a third of it on a technology stack pulling all the group's services onto one platform, and that spending sits on top of a longer run of Paratus investment in the country since 2018 (September 2025).

The public money is foreign and smaller. [China has committed CNY 98 million, about US\\$14.4 million at 2026 rates, to Namibia's Smart City pilot](#), paying for digital infrastructure, artificial intelligence capability and skills development under a decade-long programme to modernise the capital (2026). Domestically, [Cabinet approved a US\\$20 million ICT venture capital fund for SME founders, women among them](#) (September 2025), which is an approval and not yet an investment in anything.

Two commitments carry the whole of Namibia's digital-sector financing across 2025 and 2026, worth roughly US\$48 million between them once the two announcement currencies are converted, and split almost evenly between them (2026). They are different kinds of money — a foreign state's contribution to a municipal pilot, and an operator's capital spending on its own network — so the even split says nothing about who funds Namibian digital.

ICT Infrastructure

Connectivity

Namibia is one of the [few African states to have shut Starlink out](#). The Communications Regulatory Authority of Namibia [refused the company a Class Comprehensive telecommunications licence and the spectrum it needed](#), in a decision gazetted on 23 March 2026, after finding it met only three of six statutory criteria. Ownership decided it: the applicant is wholly foreign-owned against a rule requiring 51 percent Namibian ownership, and no ministerial exemption was granted; compliance history and national security made up the rest. Its chief executive Emilia Nghikembua [has said the issue has never been whether satellite technology is beneficial but whether an applicant complies with the statute](#). Starlink said it would appeal, and although CRAN upheld the refusal on 24 June 2026, a fresh application had been filed on 8 June.

Paratus broke the state-linked duopoly on 2 September 2025, launching Namibia's first privately owned mobile network, a data-only LTE and 5G service, in a market where the majority state-owned MTC held more than two million subscribers and over 90 percent of the market against TN Mobile's roughly 336,000 (September 2025). MTC switched on 5G in Windhoek, Swakopmund and Walvis Bay in August 2025. Reach beyond the towns is the constraint: 4G covered about 88 percent of the population in 2024, against a government target of full coverage by 2030, but a population density of 3.76 people per square kilometre keeps 5G phased and city-first while rural areas stay on 2G and 3G (February 2026). Twenty-five PowerCom towers costing N\$40 million were inaugurated across nine regions in August 2025, and the Equiano subsea cable lands on the coast.

Coverage now runs ahead of use. The regulator has proposed tax breaks on 4G-capable handsets, treating device affordability rather than coverage as the binding constraint, after an operator reported low use of its 4G towers in Kunene (February 2026). Internet and computer access is Namibia's most improved of all 96 IIAG indicators, up 38.1 points over 2014-2023 to 49.6 out of 100 and 13th of 54 — rapid gain that still leaves it below halfway, while mobile communications ranks only 21st of 54.

Data Storage

Namibia has no government data centre and no government cloud. [Groundwork on a National Data Centre was only to begin in the 2025/26 financial year](#), the concept note and project proposal were still being finalised in September 2025, and the country's single Tier 3 facility is privately operated. There is no shared cloud platform available to all government entities, no policy on where government cloud data may be hosted, and no monitoring or publication of government cloud usage, security or savings (2025). Nor is there a hyperscale region in the country: the nearest are Amazon Web Services' Cape Town region and Microsoft Azure's South Africa North and West (2025). Namibia imposes no data localisation requirement and does not regulate cross-border personal data flows, so government, financial and personal records may lawfully be held outside the country.

What exists commercially is one facility. Paratus's Armada data centre outside Windhoek has been operational since September 2022, with multi-carrier connectivity including the Equiano submarine cable and Trans-Kalahari fibre.

The rest is pipeline and pitch. Building a national data centre is a stated commitment of the 2025-2029 digital transformation strategy, which frames local hosting of government data as technological sovereignty, and Cabinet directed the ICT ministry and the Office of the Prime Minister in September 2025 to finalise the concept note and proposal so that national data could be hosted locally. Nine months later the ICT minister was citing Angola's sovereign data centre in Luanda as encouragement for Namibia's own ambitions (June 2026). The only build with a vendor attached is Chinese: under a smart-city pilot funded by China, Huawei is to deliver smart-city systems and a national data centre in partnership with the home affairs and ICT ministries (July 2026). Beyond that sits positioning: a green hydrogen and solar development near Walvis Bay is put forward as the power base for a prospective data-centre industry, and industry voices are arguing in the Namibian press for Namibia as an "African digital sovereign" and a natural host for AI compute on cheap renewable power and coastal cooling.

Energy

Namibia generates almost all its own electricity from renewables and reaches barely half its people with it. Renewables supplied 93.37 percent of domestic generation in 2021, principally from the Ruacana hydro plant with growing solar capacity, while only 56.7 percent of the population had access to electricity in 2023. The divide is geographic: 75.0 percent of the urban population had access in 2023 against 34.1 percent of the rural population, a gap of about 41 percentage points.

Where supply exists it is not dependable. Only around 37 percent of Namibians said in 2024 that their electricity supply is reliable, and dissatisfaction with government performance on electricity provision is rising. The weakness is not in the transmission core: the national utility reported transmission system availability of 99.81 percent for 2024/25 with no load-shedding or power outages, and 4,856 GWh into the system of which 44.6 percent was generated locally, and it inaugurated a N\$394 million digital substation outside Swakopmund in June 2026 as the main transmission supply point for the coastal network. What is missing is reach to the household. Access to energy scores 50.3 out of 100 in the 2024 IIAG, 17th of 54 and up 5.5 points over the decade — improving, but from a level that leaves rural electrification a live constraint and far below the country's standing on most other governance measures. An off-grid electrification policy is running, with deployment reaching roughly 12.6 percent of the population served off-grid (2023).

The cost to the digital estate is concrete. More than 180 rural schools have no electricity at all, and solar electrification has so far reached only around 15 of them (2024), which forecloses digital record-keeping and computer use at those schools before devices or connectivity come into the question.

Against that record sits a much larger claim about what the country could power. Namibia has one of the strongest solar resources on Earth, with over 3,000 hours of sunshine a year, and a green hydrogen and solar development near Walvis Bay is put forward as the power base for a prospective data-centre industry. That is generation potential rather than grid capacity delivered, and the two have not yet met.

Technical Capacity

Namibia runs its identity systems on other countries' technology. The Nam-X interoperability layer was built with Estonian partners, the Integrated Border Management System runs on Thales technology, and the e-ID is being prepared with UNDP support (2024) — working systems, but only partial in-country capacity to build or replace them.

The administrative machinery around the technology is the stronger half. Effective administration scores 77.2 out of 100 in the 2024 IIAG, 4th of 54 and up 10.9 points since 2014, and a ministry leads public sector digital transformation, with a cross-government forum bringing senior digital officials together on whole-of-government digital, data, technology and capacity questions (2025). But the statistical system is the weakest of Namibia's public administration measures and one of its poorest rankings in the whole index, and budgetary and financial management is among the sharpest deteriorations of the decade, and there is no government strategy or programme to improve digital skills in the public sector (2025). The software developer community outside government is small (2026).

The gap shows at the edge of the state. The police operate a central digital backend of case and biometric systems, while rural stations still open dockets, occurrence book entries and statements by hand and store them in physical cabinets, officers willing to digitise but without the funding or infrastructure to do it (2022).

Cybersecurity

Namibia's national incident response team is a unit of its communications regulator. NAM-CSIRT sits inside CRAN rather than in a standalone cyber agency, and its national threat monitoring runs on CRAN's own sensors; the same regulator states that it acts as Namibia's root certification authority, anchoring the national digital trust framework and regulating certification-service providers. NAM-CSIRT's own count for 2025 was 1.7 million cyberattacks against Namibian targets, 37 percent more than in 2024, alongside 2.23 million detected vulnerabilities, and its April-June 2026 quarter logged 513,921 vulnerabilities and 161,547 security events, up 39.8 and 56.7 percent on the previous quarter, with internet-exposed systems named as the principal cause.

The law behind that machinery is thin. Namibia has neither a cybercrime statute nor a data protection law: both bills were still being finalised as at July 2026, and enforcement in the interim rests on the Electronic Transactions Act, the Communications Act and the Penal Code. A Cybercrime Bill that would designate and protect critical information infrastructure was still in draft in January 2026. The one binding incident-reporting duty is sectoral: Bank of Namibia determination PSD-12, in force from 1 July 2023, requires national payment system participants to report a successful cyberattack to the central bank within 24 hours of detection and to submit a full impact assessment covering financial, data and availability loss within a month.

The losses are not hypothetical. Telecom Namibia lost data to the Hunters International ransomware group in December 2024, exposing records belonging to ministries, regional councils and municipalities. Namibia scored 37.93 out of 100 in the ITU's Global Cybersecurity Index 2024, in the index's lowest tier, classified there as "evolving". Cabinet approved proposals in November 2025 for a government-sector CSIRT, security operations centres and cyber-skills training for public servants, coordinated by the Office of the Prime Minister. That office names internal misuse of e-ID data as a larger risk than external attack.

DPI

Data Exchange

Namibia's government data exchange has been in service since 2014 and still does not reach the whole of government: Nam-X, built on the Estonian X-Road stack, connects civil registration, the national population register and some government databases, but full inter-agency coverage has not been achieved. What crosses it is mostly identity. The platform links the population register to e-birth and e-death notification, to passports and to the statistics system, while memoranda of understanding to bring in the ministries responsible for gender and child welfare and for finance, the Electoral Commission and the poverty eradication programme were still being finalised in 2025, and its most attested use is verification: it connects the population register to other public and private

registries so that an identity can be checked, and was used to validate applicants for the COVID-19 emergency income grant. Nam-X is run from the Office of the Prime Minister, with regional points of presence giving decentralised access to services.

Exchange beyond identity happens with very little governing it. The financial management, tax, customs and social insurance systems each pass data to other government systems, yet the same 2025 assessment records no government interoperability framework, no data quality framework, no guidance for replacing legacy systems and no monitoring of whether government systems are up, and no law mandates cross-government data sharing.

The absences are concrete. The immovable property registry and the cadastre are interoperable neither with each other nor with other public services; the automated exchange between the beneficiary registry and the wider platform is not operational; and the revenue agency and the business registry agreed in February 2024 to join tax and company records, a connection still being designed rather than built.

Digital Identity and CRVS

What limits identity coverage is collection, not production: the Ministry of Home Affairs printed 166,237 national identity documents in the fifteen months to 30 June 2026 and 125,999 were collected.

About 40,200 of those documents were never handed over in that period to 30 June 2026, roughly 24 per cent of what was produced.

At the 2023 census 85.4% of Namibians aged 16 and over held a national identity card, and citizens' sense of how easily an identity document can be obtained fell 29.9 points over the decade to 2023, to 37.3 out of 100. Campaigners petitioned in July 2026 to hold the electronic ID back until the document backlog is cleared.

The foundation is clean. Civil registration and identity management run on one system, so vital events update the identity record through a single unique identity number and cards are issued out of the population register; enrolment captures all ten fingerprints and a facial photograph; and it covers citizens and permanent residents, compulsory from 16. Every mobile subscriber had to register one, or a passport, by 31 December 2023 or lose the SIM, and payments to more than 15,000 old-age pensioners were suspended in 2026 when their identities could not be matched against the register.

The electronic ID is scheduled, not issued. Cabinet approved it in 2023 and the rollout was carried in the 2026/27 budget, the launch has slipped from July to September 2026, and the current card cannot be used for digital or remote authentication. The identity statute governs it: the minister told the National Assembly in June 2026 that implementation will comply with the Civil Registration and Identification Act 2024, under which chip use is not compulsory and disclosure is consent-based, with further safeguards left to data protection legislation that has not been enacted. The Office of the Prime Minister names internal misuse of e-ID data as a larger risk than external attack. The certification machinery behind the card was contracted at the end of August 2026: the communications regulator and a German identity vendor signed an implementation agreement connecting the ministry to the national trust framework, the ministry becoming the first certification service provider through its issue of electronic identity documents. That is a signature, not an issued card.

Omusati residents protested in June 2026 on the claim that the card would microchip and track them, and by August the ministry was rebutting fresh claims that it exposes citizens' data, with its awareness campaign out in Omaheke.

Digital Payments and Fintech

Namibia's instant payment rail went live in June 2026 with the central bank as its operator rather than only its supervisor: the Bank of Namibia runs the Instant Payment Solution through a subsidiary, Instant Payments Namibia, with India's NPCI deploying the system inside the central bank's own environment. It became operational in June 2026 after a live pilot, with Bank Windhoek, Nampost and Letshego Bank Namibia the first participants. It is carried publicly as WayaMe, customers transacting through their existing bank applications, e-money wallets or USSD services while the payment is routed through the national instant-payment infrastructure, and it is designed to carry bank-to-bank, bank-to-wallet, wallet-to-bank and wallet-to-wallet payments between participating institutions.

Government payments moved first. The first cohort of marginalised beneficiaries was paid over the new rail on 17 June 2026 through Bank Windhoek, with pensioners and other grant recipients to follow, but only the first phase of grant categories has migrated: salaries, pensions and the remaining social grants have not. Across borders, Namclear agreed in March 2026 to connect the country to TCIB, the SADC regional instant cross-border scheme, with low-value push payments to Common Monetary Area countries introduced in phases. Bank-to-bank transfers already run on NamPay, the upgraded national electronic funds transfer system, whose near-real-time credit stream carries person-to-person, salary and supplier payments.

The rules exist and are largely unpublished. The Payment System Management Act 14 of 2023 gives the Bank of Namibia power to approve payment system rules and membership criteria and carries a dedicated part on consumer protection; a determination effective 15 February 2024 licenses non-bank payment service providers and requires audited annual accounts that are not published; and operators must submit scheme rules on governance, access, risk, settlement finality and fees for approval, on forms marked confidential, with no pro-poor or consumer-representation requirement imposed. A 2025 determination bars charging customers for erroneous transactions, requires reversal within 14 days and sets complaint-handling timelines, and the central bank concedes that transaction costs are a live grievance needing industry-wide action.

Access to banking scores 45.9 out of 100 in the 2024 IIAG, among Namibia's ten weakest measures and its ten most improved, and still ranks 6th of 54. Fraud is climbing against that base: the Bank of Namibia put annual banking fraud losses at N\$73.9 million in 2025, against roughly N\$8.7 million in 2020.

Registries

Civil registration is one of only two measures on which Namibia scores a perfect 100.0 out of 100 in the 2024 IIAG, ranking 1st of 54 and unchanged across the decade to 2023. Completeness is another matter: birth registration fell to 62.0% in 2021, from 89.6% the year before.

The machinery behind the score is real. The National Population Registration System links 62 decentralised registration centres for real-time recording of births, deaths and marriages, with certificates issued to the applicant on the spot, and electronic birth and death notification now reaches 153 health facilities, private hospitals, police mortuaries and old age homes feeding directly into the register.

Other registers are further back. Company registration is only partly digital: name reservation, an application-status check and a login portal are online, but registration still requires notarised memoranda and articles of association lodged in person. The Deeds Registry remains substantially paper-based while a computerised system is implemented, and the World Bank's 2025 assessment found no digital title certificates, no digital cadastral plans, no public online identity-checking database and no interoperability between the property registry, the cadastre and other public services. Most identified communal land rights have been registered under the national communal land administration system.

The registers that pay people are the least joined up. An integrated beneficiary registry and an upgraded social assistance database were both completed in 2022, are still not fully deployed across all schemes, and the automated exchange with the wider government platform that was to follow is not operational, while the Integrated Social Assistance System, linked to the population register, is in active use, verifying beneficiaries' identity and life status and paying them under a March 2026 Cabinet decision on the Conditional Basic Income Grant. Coverage across the individual schemes is highly uneven.

The voters register is the outlier. The Electoral Commission has run a fully digital biometric roll since replacing paper between 2010 and 2014, and used an upgraded mobile registration system for the 2024 general registration of voters, signing up around 91% of an estimated eligible population before the November 2024 elections.

The revenue register is the one that has grown. The revenue agency counted 1.0 million registered taxpayers at 31 March 2025, of whom 575,136 were registered as electronic filers, on 1.04 million active tax accounts, and a trader identification number became mandatory on every import and export declaration on the customs platform from 1 April 2026, tying the customs register to the taxpayer register.

Addressing is the register that does not exist. The postal operator regrouped its codes from postal administrative regions into political regions and states that this phase covers sorting and delivery only, not administrative or geographic addressing, with a physical-address phase still to come, while development of a national addressing system is named as an objective of an April 2026 memorandum between the communications regulator and the statistics agency, with no specification, budget or date given.

Sectoral management information systems

Only 18 clinics nationally were digitally connected as of Namibia's 2025/26 health budget. Rural primary care is overwhelmingly paper at the point of care, with nurses recording on patient cards and health passports and the data aggregated into the national DHIS2 system only at district or regional level — the pattern a national digital health policy launched in February 2026 gives itself a decade to change.

Education runs the same way. Rural primary schools keep attendance, enrolment and outcomes on paper registers, submitted twice yearly on physical forms to regional offices where the ministry's EMIS division digitises them centrally; the most recent census counted 2,036 schools as of January 2024. Policing is split down the middle: the police have run a central digital backend since 2010, holding case dockets and crime statistics across 16 linked databases, with a biometric identification system added in 2016, while rural stations still open dockets, occurrence book entries and statements by hand and store them in cabinets.

The finance systems are real but narrow. The financial management information system runs on off-the-shelf software, covers central government only and supports treasury execution rather than the full public financial management cycle, and sits beside a debt management system but no public investment management system, so no project database and no published investment results. Budgetary and financial management is Namibia's fifth most deteriorated measure in the 2024 IIAG, down 30.5 points over the decade to 55.6 out of 100. Revenue is the exception: the tax agency's Integrated Tax Administration System offers round-the-clock self-service registration and filing, and customs runs on off-the-shelf software that exchanges data with other government systems.

None of the five core administrative systems in operation — financial management, the treasury single account, tax, customs and e-procurement — has governance arrangements covering compliance, security and audit trails.

The gap at the centre is the workforce: there is no human resources management information system with self-service for public employees and managers and no payroll system linked to one, so no public service HR data is exchanged and the national ID is not used as an HR identifier.

Other GovTech and e-Gov

Namibia's online public service portal is informational — publishing information rather than completing transactions — though it offers omnichannel access and a business can be started through it. Transactional services beyond that are confined to a handful, and the national digital strategy's target of a portal carrying ten essential services is not operational.

Of the 32 core public sector digital capabilities the GovTech Maturity Index checks for in Namibia in 2025, 15 are in place, 16 are absent, and one — a whole-of-government approach to digital transformation — is in draft.

What does transact sits on separate systems. The tax portal takes registration and filing with omnichannel access, the social insurance and pension portal takes registration and benefits transactions, and the e-procurement portal publishes tender notices and contracts, though not to the Open Contracting Data Standard and without exchanging data with other government systems. Driving licence renewals and test bookings are online.

The gaps run in the citizen-facing direction. There is no government job portal, and no mobile app for public services, no e-residency, no involvement of citizens or businesses in designing e-services and nothing published on service delivery performance or user experience.

The institutions are further ahead than the services. A ministry leads public sector digital transformation and a cross-government forum brings senior digital officials together; an established GovTech entity and a coordinating body exist, but nothing monitors digital spending across government and the institution publishes no annual progress report, and no GovTech or digital transformation strategy has been adopted — against a stated target of digitising all government services by 2029/30. Effective administration scores 77.2 out of 100 in the 2024 IIAG, 4th of 54 and up 10.9 points since 2014, while public administration as a whole fell 8.4 points over the same decade.

Donor money has gone to the legislature, not the executive: the digital components of the EU's indicative programme fund an e-records database of parliamentary debates, an online library and research service, an e-consultation system and members' e-skills, within a EUR 2.5 million governance envelope of an indicative EUR 37 million covering 2021 to 2024.

Digitalisation

Digitalisation of sub-national government

Namibia's sub-national digital state is being built from the centre rather than by the tier it serves. Cabinet directed in September 2025 that a legal framework be developed for a Windhoek smart city, working with the Ministry of Urban and Rural Development — a framework to be written, ahead of anything a municipality would run.

The public argument about it has been about control of the data rather than the build. On 4 August 2026 Rodney Cloete, the Independent Patriots for Change's shadow minister of international relations and trade, argued that Namibia should not accept a Chinese smart-city grant, which he put at N\$245 million, without enforceable safeguards, resting the case on the country having no data-protection statute in force to bind a foreign supplier.

Where national systems do reach past the capital, they stay national: Nam-X is managed from the Office of the Prime Minister, with regional points of presence giving access to government services outside Windhoek (2024).

Rural digital data capture

At the point where rural Namibia meets the state, the record is still made by hand: a 2022 study of the Omusati region found police officers opening case dockets, occurrence book entries, affidavits and statement forms on paper at the scene and filing them in physical cabinets, willing to digitise but with neither the funding nor the infrastructure

to do it. The same shape holds in the two services that reach furthest into the countryside. Rural clinics [record on patient cards and health passports at the point of care](#), the data entering the national health information system only once it has travelled to district or regional level, and rural primary schools [keep paper attendance and enrolment registers, sent twice yearly on physical forms to regional education offices for the ministry to key in](#). Digital capture in rural Namibia is, for the most part, transcription performed well upstream of the place the data was created.

Public money is going to the service points rather than to the paperwork. The Universal Service Fund's first priority is [the five regions where 4G coverage falls below the National Broadband Policy's 80 percent threshold – Kavango West, Kavango East, Kunene, Ohangwena and Oshikoto – with 16 schools and four clinics selected for uncapped Wi-Fi \(March 2025\)](#). That leaves the older constraints in place beneath it, [the rural half of the country's electricity access gap and the rural schools still without power at all](#), which decide whether a school or clinic can operate a system rather than merely reach one.

What is captured about rural Namibia is thin at the national level too. The country carries [no score on any of the four rural-economy indicators in the 2024 Ibrahim Index \(2023\)](#), and its statistical programme has [gone without an agricultural survey for a decade](#) while running labour surveys repeatedly.

Technology

AI

Artificial intelligence reached Namibian public life as a fraud before it reached it as a service: in September 2025 a [deepfake video circulated showing the Prime Minister, her deputy and ICT minister Emma Theofelus appearing to endorse an investment scheme promising N\\$35,000 a week](#), which Theofelus disavowed, and in November [the Presidency denounced a viral deepfake audio and video impersonating the President](#). No prosecution, takedown or regulatory action followed on either. By 2025 NAM-CSIRT had [catalogued a wider set of AI-driven frauds running against Namibians](#), running from deepfake investment scams through double-extortion ransomware against municipalities to Telegram-based phishing and smishing that impersonates delivery services.

On the other side of the ledger there is a prospectus and very little running. Namibia had [no national strategy for disruptive or innovative technologies](#) in 2025, no artificial intelligence policy had been adopted as at August 2026, and [no artificial intelligence was integrated into the national data exchange](#) that year. What exists is positioning: Huawei is to deliver an "AI-ready" [national data centre](#) under a Chinese-funded smart-city pilot, the President [pressed Huawei at its Shenzhen headquarters in July 2026 to create AI jobs and skills in Namibia](#), and an industry case is being argued in the Namibian press for the country as a host for AI data centres on cheap renewable power and coastal cooling (February 2026). Machinery for writing the strategy does now exist: [the research commission formally commenced its development in September 2025 under the education ministry's direction, appointing technical advisory committees of artificial-intelligence experts to lead the process](#). A vendor label, a siting thesis and a drafting committee are not a system in operation.

ICT Industry

Namibia ranks [4th of 54 African states for its business and labour environment in the 2024 IIAg, on 58.7, but only 27th for business and competition regulations, which scored 54.5 after a 10.8-point fall over the decade to 2023](#): a strong overall standing rests on a weakest component that governs who may enter a market at all. The market itself is small by construction: [2.6 million people and GDP per capita of USD 4,742.8 \(2023\)](#) support few firms of scale, and mobile remains [dominated by the majority state-owned MTC, with the first privately owned network launching only in September 2025](#), built by Paratus with Nokia as network partner and Cerillion as digital technology partner.

There is no structured channel by which the state buys from domestic small firms. As at 2025 Namibia had [no policy to support GovTech startups or private sector investment in government technology](#), [no public financing to startups or SMEs for innovation](#), and [no procurement policy prioritising bids from startups and SMEs](#), and published nothing on progress in the area.

Innovation ecosystem

The largest venture round raised by a Namibian startup is [JABU's US\\$3.2 million for B2B e-commerce and retail distribution in 2022](#), which sets the ceiling of a thinly funded startup sector; the country ranked [102nd of 133 economies in WIPO's Global Innovation Index 2024](#). At least four innovation hubs are active — [ScaleUp Namibia at the Namibia Investment Promotion and Development Board](#), [Impact Tank](#), the [Afrisource Innovation Center](#) and [RLabs Namibia \(2026\)](#) — but they operate without any dedicated statute behind them: Namibia has no start-up act, and none had been gazetted or tabled as at August 2026.

Inside government, innovation is an institution with no public record of what it does: an [established public sector innovation entity](#) and a [public sector innovation strategy](#) exist, with [no collaboration with the private sector](#) and [neither the institution nor its programmes publishing performance or results \(2025\)](#). Cabinet approved an [ICT venture capital fund aimed at SME founders, with women founders named as a target group](#), in September 2025.

Capacity

Literacy

[Equality in education ranks 34th of 54 African countries and has slid over the decade to 2023](#), one of Namibia's ten worst governance measures in a country that sits 6th on the continent overall. The weakness is not in the teaching workforce, which is [one of the ten best measures Namibia has, 6th of 54 and sharply improved since 2014](#), while the [quality of the education it delivers ranks 13th](#). Nor is it enrolment, [the strongest-moving part of an education score that ranked 12th of 54 in 2023 and was otherwise flat across the decade, with completion also up](#). What the country has built is a school system that reaches and staffs itself well and distributes what it teaches unevenly.

The floor beneath any digital system is still a paper one. Rural primary schools [keep attendance, enrolment and learning outcomes on paper registers](#) and send them twice yearly on forms to regional offices for the ministry to digitise centrally (January 2024), and [many rural schools have no electricity at all](#), which alongside thin rural mobile coverage and few devices keeps computers out of those classrooms (2024). [No government digital skills programme reaches schools or citizens \(2025\)](#), so a pupil's exposure to the systems the state is digitising depends on where the school is.

Training and skills

The World Bank's 2025 GovTech assessment found Namibia with [no government strategy or programme to raise digital skills in the public service](#), [none for citizens or schools](#), and [no published results in the area](#). What stands in its place is a run of agreements. The Namibia Institute of Public Administration and Management [signed a memorandum of understanding with the Global Cybersecurity Centre in June 2026 to train public servants in cybersecurity](#), after Cabinet [approved proposals in November 2025 that included cyber-skills training for public servants](#), and in July 2026 President Netumbo Nandi-Ndaitwah [pressed Huawei to create AI jobs and skills in Namibia](#). Each is an intention to train rather than training under way, and all three point at the same narrow group: public servants and the cyber and AI functions the state is standing up. Teaching that is actually running sits outside government, in the tertiary institutions and in the country's [innovation hubs \(2026\)](#).

Research institutions

Knowledge production in this field is concentrated in the tertiary sector rather than in government or a national institute, with the Namibia University of Science and Technology [hosting several UNESCO research chairs](#) (2024).

Inclusion

Access to services

Namibia's oldest cash transfer reaches almost everyone it targets and its newest a seventh of them: the [Old Age Grant](#) covered about 97% of its target population, the [Disability and Vulnerable Children's grants](#) about 75% each, and the [Conditional Basic Income Grant](#) about 14% of targeted households (2024).

What separates them at the counter is a document. A national identity document is mandatory for a grant, and beneficiaries are re-verified against the population register — pensioners whose identities could not be matched had payments stopped until they produced one (2026). The constraint sits at handover rather than production: much of what the ministry printed in the fifteen months to June 2026 was [never collected](#), and [Namibians' reported experience of obtaining an identity document has deteriorated sharply over the decade](#) (2023).

Eligibility is drawn by status too: [permanent residents hold pink identity cards and refugees yellow ones, while non-nationals staying less than a year, including temporary work-permit holders, cannot hold one at all](#) (2024). Refugees reach digital money by another route — [UNHCR paid cash assistance to around 6,000 people at the Osire settlement through MTC's mobile money service](#) (2025), as humanitarian aid rather than ordinary account access.

On payments, [government grants began moving onto the national instant payment system in June 2026](#), and the Bank of Namibia has [conceded that transaction costs are a live grievance needing industry-wide action and reset its own objective from access to effective use, naming cash reliance, thin rural provision, informal-sector barriers and the digital divide as what remains](#) (July 2026).

The state is meanwhile buying connectivity for the places services are delivered from: the [Universal Service Fund](#) launched in March 2025 with [N\\$145 million of government seed funding over three years](#), with [N\\$31 million already gone to MTC for nine network sites](#), and in January 2026 the ICT minister [committed N\\$80 million to seven years of free internet for schools and clinics](#). What a citizen can transact remains thin: the public service portal publishes information rather than completing transactions, and there is [no government channel for complaints about service delivery and no accessibility support for disabled users](#) (2025).

Digital divides

Inclusion and equality is one of [Namibia's three worst-scoring governance sub-categories](#), at 58.3 out of 100 in 2023 and down 4.5 points over the decade to 16th of 54 African states, and the [equal access to public services measure](#) inside it is lower still at 53.0 and also falling. Both sit far below the country's 6th place on the continent overall.

The divide runs on geography before anything else. [Namibia's population density is among the world's lowest](#), which makes towers and backhaul expensive and has left the 5G rollout city-first while rural areas stay on 2G and 3G (February 2026). Rural households are [far less likely than urban ones to have electricity](#) (2023) and [many rural schools have none at all](#) (2024), so in the north and north-west the obstacle to going online is as often power as signal. The [Universal Service Fund's priority regions](#) — Kavango West, Kavango East, Kunene, Ohangwena and Oshikoto — are those where 4G coverage falls short of the national broadband target (March 2025).

Measured at institutions rather than at households the picture is worse: [many of Namibia's schools, and a smaller share of its health facilities, had no 4G service at all](#) (2022), a gap much wider than any headline population figure and one that falls on the places people go to be served.

Where the network does reach, what limits use is the handset rather than the mast. MTC has reported that [much of its 4G capacity in the Kunene region goes unused](#) (February 2026), coverage built out ahead of the devices needed to take it up, and CRAN has [proposed tax breaks on 4G-capable handsets](#) (February 2026), treating affordability rather than reach as what keeps Namibians offline. [Mobile data is expensive by international comparison](#) (2024), which bites on the same households.

Data

National statistics

The statistical system is the weakest part of Namibia's public administration and one of its weakest rankings anywhere in the Ibrahim Index: [50.9 out of 100 and 25th of 54 in 2023, after a gain of 4.4 points over the decade, in a country that ranks 6th on the continent for governance overall](#).

The gap is in collection rather than in output. Namibia has run [no agricultural survey and no business survey in the last ten years, and only one health survey and one household survey in that period, against three or more labour surveys](#) (2023), even though [all three census types the World Bank tracks — population, agricultural and business — were conducted within the same decade](#). Across the five pillars of the same assessment, [data products score in the 70-89% band and data use, services and infrastructure in the 50-69% band, while data sources — censuses, surveys, and administrative and geospatial collection — fall to the 20-49% band](#) (2023). The published series hold up better than the machinery feeding them.

The first published rules for that machinery arrived in September 2025, when [the statistics agency issued five national statistics system standards: statistical quality indicators, statistical metadata, leave no one behind, data and statistics presentation, and data and statistics dissemination](#). They rest on the Statistics Act 2011, which constitutes the agency and provides for the national statistics system and the spatial data infrastructure. What they govern is how statistics are described and published, not how they are collected, which is where the assessed weakness sits.

That machinery is largely paper at the point of contact. Rural primary schools [record attendance, enrolment and learning outcomes on paper registers and send them twice a year on physical forms to regional education offices, where the ministry's EMIS division digitises them centrally for the annual education census](#) (January 2024), and rural clinics [remain overwhelmingly paper-based at the point of care, with nurses writing on patient cards and health passports and the data reaching the national DHIS2 only at district or regional level](#) (2025). Vital statistics show what that costs: [birth registration completeness fell sharply in 2021, on the statistics agency's own reporting](#).

Nor does anyone in government own the problem: [no dedicated entity is responsible for data governance or data management, and there is no data governance strategy or policy](#) (2025). Financial statistics are the standing exception — the Bank of Namibia's [national payment system figures appear on a regular quarterly cycle, each edition within days of quarter-end](#) (2026).

Open data

Namibians have a statutory right to information — the [Access to Information Act 8 of 2022](#) — and the law is [rated effective with an entity monitoring its implementation; what is missing is any published account of how it is being applied, down to the number of information requests received and granted](#) (2025). The right exists and its exercise is invisible.

Access to public records is nonetheless among the areas Namibia has improved most: [accessibility of public records rose 15.4 points over 2014-2023 to 54.6 out of 100, ranking 5th of 54, and disclosure rose 8.9 points to 53.4, ranking 6th \(2023\)](#) — top-six placings resting on scores barely above the halfway mark, which says as much about the continent as about Namibia.

What is open is what concerns money. On the 2024 Open Data Inventory, [government finance scores 7 out of 10, balance of payments, international trade and price statistics 6 each, and national accounts 5.5, against 4.5 for labour and for money and banking](#), while the social statistics are markedly less open: [education facilities and reproductive health score 2, education and health outcomes 2.5, gender statistics 3 and poverty 4.5, with only population and vital statistics reaching 5.5.](#)

No category in that 2024 profile scores above 7 out of 10, the mean across the 19 scored categories is 4.2, and 13 of the 19 sit at or below 4.5 — an open data estate that is uniformly mid-to-low rather than strong in parts.

Publication, where it happens, is manual and unstandardised. The [national open data portal carries only basic information and datasets and is updated mostly by hand rather than dynamically through APIs \(2025\)](#), and the [e-procurement portal, which does publish tender notices and contracts, does not publish them to the Open Contracting Data Standard and exchanges no data with other government systems \(2025\).](#)

On all 16 transparency measures the 2025 GovTech assessment applies — results, progress or spending for everything from cloud, data protection and digital transformation to right to information, online services and digital skills — Namibia is recorded as publishing nothing.

Use of satellite data

Namibia's one working piece of Earth observation capacity arrived as a gift: a Chinese-supplied satellite ground station receiving remote-sensing data was [handed over at the Telecom Earth Station outside Windhoek in February 2026 and is operated by trained Namibian staff](#), with control of the data it takes down already raised in Parliament. It is a donated facility rather than a national programme, and geospatial collection sits inside [the weakest of the five pillars of Namibia's statistical performance \(2023\)](#), alongside censuses, surveys and administrative data.

Satellite service of the other kind is contested rather than absent. The regulator [refused Starlink a licence in March 2026 and upheld that refusal in June](#), having [approved a licence transfer for another satellite operator at the same meeting](#), its chief executive putting the question as one of compliance with the statute rather than of whether satellite technology is beneficial.

Geopolitics

US / hyperscaler activities

No United States government programme and no American platform or hyperscaler held an operating position in Namibia's digital infrastructure as at August 2026.

China activities

The Chinese footprint now runs to a named vendor holding the core of the state's own build: [Huawei is to deliver smart-city systems and a national data centre with the ministries responsible for home affairs and for information and communication technology \(July 2026\)](#). It rests on a [Chinese grant signed at the end of June, inside the Windhoek People-Centred Smart City Strategy, a decade-long modernisation programme running to 2036](#). Both the

money and the data centre are at the commitment stage. The one Chinese-supplied asset already in service is a [satellite ground station handed over at the Telecom Earth Station outside Windhoek in February 2026](#), receiving remote-sensing data and run by a trained Namibian team.

Sovereignty is the frame both sides have chosen, and they mean opposite things by it. [President Nandi-Ndaitwah pressed Huawei at its Shenzhen headquarters in July 2026 for AI jobs and skills in Namibia, and the company pitched the partnership as strengthening Namibian data sovereignty](#). Weeks later an opposition shadow minister argued that the smart-city grant should not be accepted without enforceable safeguards, because no data-protection statute is in force to bind a foreign supplier (August 2026), and [asked parliament for the agreements, annexures, contractors, data architecture and lifetime operating costs, for public hearings and independent security assessments, and for a Data Protection Act enacted before the build proceeds](#).

EU activities

The European footprint sits in the identity stack, and it is bilateral and commercial rather than an EU programme. [Namibia runs its identity systems with only partial in-country technical capacity: the Nam-X interoperability layer was built with Estonian partners and the Integrated Border Management System runs on Thales technology \(2024\)](#). That relationship is still growing: [an Estonian-funded project is building further Nam-X services, while memoranda of understanding to connect additional agencies — the ministries responsible for gender and child welfare and for finance, the Electoral Commission and the poverty eradication programme — were still being finalised in 2025](#).

The EU's own money goes somewhere else entirely. [The digital components of the Multi-Annual Indicative Programme for Namibia fund parliamentary rather than regulatory work — an e-records database of debates, an online library and research service, an e-consultation system and members' e-skills — within a governance envelope of EUR 2.5 million out of an indicative EUR 37 million covering 2021 to 2024](#). The same programme [names data protection once and only conditionally, among the governance sector's priorities as "possibly data protection and cyber security", and attaches no allocation to it](#).

Gulf/UAE activities

No Gulf state and no Gulf-based company held a position in Namibia's digital infrastructure as at August 2026.

India activities

India's footprint is inside the central bank. [Namibia's instant payment system is operated by the Bank of Namibia through a subsidiary, with India's NPCI deploying the solution inside the bank's own environment \(2025\) — a running system inside a core state institution rather than a plan for one](#).

The rest of the relationship is at the paper stage. [Namibia and India signed a cybersecurity memorandum of understanding at State House in July 2025](#), agreed between President Nandi-Ndaitwah and Prime Minister Modi alongside cooperation on UPI-based payments and digital governance.

ABOUT THIS DOCUMENT

EDITION	2026-09-06
CURRENT EDITION	https://corpus.data-landscapers.io/reports/NAM/NAM-status.html
THIS FILE	https://corpus.data-landscapers.io/reports/NAM/NAM-status-2026-09-06.pdf
LICENCE	CC BY 4.0

Figures are dated because most are time-varying: a figure carries the date it was true, not the date you are reading it. Where the base holds no reliable statement, the document says **Not held** rather than leaving a silence — those are counted, and listed at the end.

This is a dated edition and is not revised after publication. If a figure here has moved, the current edition will say so.